AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M109
.Net Programming

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 3 /Dec. /2025

Number of Exam Pages: Five (including this cover sheet)	**Time Allowed:** 90 minutes

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are not allowed.

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Part 1: Multiple Choice Questions (MCQ) [10 marks]

1- When using a StreamWriter, what happens if you omit the Close() method?.

a) Data is automatically saved
b) Data may not be written to the file
c) The file becomes read-only
d) An exception is thrown immediately

2- Which statement about CheckBoxcontrols is TRUE?

a) Only one CheckBox can be selected at a time
b) CheckBoxes support independent selection
c) CheckBoxes must be placed in a GroupBox to function
d) CheckBoxes do not have a Checked property

3- What happens when you assign a decimal to an int without casting?

a) Automatic rounding occurs
b) Compilation error
c) The value is truncated
d) Overflow exception

4- What is the purpose of the TryParse() method?

a) To convert strings without throwing exceptions
b) To test mathematical expressions
c) To validate file paths
d) To compare two strings

5- The …….. statement, when executed in a for loop, skips the current iteration and begins the execution of the next iteration of the loop.

a) Next
b) Skip
c) Continue
d) Break

6- When you create a new project using Visual Studio, it will be stored in a _____ at the location you specify on your computer's disk.

a) Solution folder
b) Solution file
c) Project folder
d) System fileControls

7- In object-oriented programming with a GUI interface, a window displayed on the screen is called a _____.

a) Window Object
b) Form Object
c) Screen Object
d) Frame Object

8- What will be the output of the following code: int i=3; Console.Write(i++ + ++i);

a) 7
b) 8
c) 5
d) None od the mentioned

9- A double data type can store larger numbers than a decimal data type.

a) True
b) False

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10- A program with a logic error can be compiled and run, but it will produce incorrect results.

a) True b) False

Part 2: Short Questions [20 marks]

Q1- Explain through examples the difference between value and reference data types in C#.

Q2- What is the output of the following program? And what will be the output if we replace the shaded word with "break"?

```
int count = 0;
while (count < 5)
{
   count++;
   MessageBox.Show(System.Convert.ToString(count));
   count++;
   if (count == 3) continue;
}
MessageBox.Show("Final: " + count);
```

Q3- The following code includes 7 errors (an error per line). Find the errors and correct them.

```
int x, y, my Num = 0;
double d = 10 + x; x = 3;
y = d + 2;
if (x > 5) ;
Console.WriteLine("Result1 = {0}", x);
Else
Console.WriteLine("Result2 = {0}", myNum);
```

Q4- Trace the following code. Specify the result of each line, not only the final output.

```
static void Main(string[] args)
{
   int x = 20;
   int y = x / 3;
   double z = Math.Pow(2, 3);
   Console.WriteLine($"Initial: x={x}, y={y}, z={z}");
   for (int j = 1; j <= 3; j++)
   {
      if (j % 2 == 0)
      {
         x -= 3;
```

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```
            y += j;
        }
        else
        {
            x += 4;
            z = x / 4.0;
        }
        Console.WriteLine($"Iteration {j}: x={x}, y={y}, z={z}");
    }
    try
    {
        Console.WriteLine("\nFinal Calculation:");
        Console.WriteLine(x / (y - 6));
    }
    catch (Exception ex)
    {
        Console.WriteLine(ex.Message);
    }
}
```

Q5- Complete the following paragraph with the suitable terms.
[syntax error, break, exception, logic error, run, Text, decimal, try-catch ,constant, selected, checked, const]

1- A(n) _____ does not prevent the program from running, but causes it to produce incorrect results.
2- When you start a program, it enters _____ mode.
3- The _____ property of a TextBox control holds the text entered by the user.
4- To handle exceptions, you can enclose code in a _____ block.

Part 3: Problem Solving Questions [30 marks]

Q1- You are tasked with creating a Windows Forms application that converts distances between Miles and Kilometers. [15 marks]

Requirements:

- The application should have a TextBox for entering the distance to convert.
- There should be two RadioButton controls:
- "Miles to Kilometers" (default selected)
- "Kilometers to Miles"
- A "Convert" button that performs the conversion based on the selected RadioButton and displays the result in a Label (e.g., "16.09 km").

The application must include:

- Input validation using TryParse() to handle non-numeric input.
- A "Reset" button that clears the TextBox, Label, and resets the RadioButton selection.

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Formulas (use these fixed rates):

- •Miles to Kilometers: Kilometers = Miles * 1.60934
- •Kilometers to Miles: Miles = Kilometers / 1.60934

Q2- Write a C# program using Windows Forms that:

- Reads a list of integer numbers (one per line) from a file named **data.txt**.
- Calculates the sum, average, minimum, and maximum values from the numbers in the file.
- Writes these statistics to a file named **statistics.txt** in the following format:

```
Sum: [calculated sum]
Average: [calculated average]
Minimum: [minimum value]
Maximum: [maximum value]
```

- Your program should include a Button labeled "Analyze Numbers" to trigger the process.

- Ensure your program correctly handles file I/O, gracefully ignores any lines that are not valid integers, and provides feedback to the user (e.g., a MessageBox indicating completion or if the file was not found). **[15 Marks]**

End of questions

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